20 TYPE CONVERSION AND CASTING • THEORY

03 dynamic cast and rtti rules

Detailed Theory, Exact C++ Rules, Memory Models & Exam Reference

03 — `dynamic_cast` Failure Behavior: Pointer vs. Reference

When a downcast `dynamic_cast<Derived*>(base_ptr)` fails at runtime (because `base_ptr` does not actually point to a `Derived` object):

- **Pointer** `dynamic_cast<Derived*>(base_ptr)`: Returns `nullptr`.
- **Reference** `dynamic_cast<Derived&>(base_ref)`: Throws `std::bad_cast` (`#include <typeinfo>`) because there is no such thing as a null reference!